**Department of Chemical & Biomolecular Engineering**
**North Carolina State University**

**CHE 330** **Spring 2011**

# Continuous Variables & Probability Distributions[†]

## Continuous Variables

Continuous variables are variables that take on fractional values, examples include temperature, pressure and pH readings, length, volume, weight and density measurements, mole number and concentration, distance travelled, lifetime, etc. A continuous variable can theoretically carry as many significant figures as one can imagine and, as such, no two continuous variables, even if they are of the same kind, can ever take on *exactly* the same value. Thus, a temperature reading of 25.0 $^{o}$C is, in principle, different from a reading of 25.01 $^{o}$C that is, in turn, different from 25.001 $^{o}$C, even though the differences may be of little practical significance. Because a continuous variable can take on an infinite number of possible values within any given interval (by simply extending the number of significant figures beyond the decimal point), the probability of the continuous variable taking on any specific value is one out of infinite possibilities or zero. It follows that a continuous variable has a zero probability of taking on any specific value. A non-zero probability can only be assigned to a continuous variable if an interval or a range of values is specified for the continuous variable.

The probability distribution for a continuous variable $x$ is given by the probability *density* function *f(x)*, which is defined as probability per unit interval of a possible range of $x$ value. The following relationships hold for a probability density function *f(x)* for a continuous variable $x$:

$$P(x_1 < x < x_2) \equiv \text{Probability of } x \text{ falling in the } x_1 - x_2 \text{ range} = \int_{x_1}^{x_2} f(x)dx \tag{1}$$

$$\int_{-\infty}^{\infty} f(x)dx = 1.0 \quad \text{for a normalized probability density function } f(x) \tag{2}$$

$$E(x) \equiv < x > = \int_{-\infty}^{\infty} xf(x)dx \tag{3}$$

$$E[y(x)] \equiv < y(x) > = \int_{-\infty}^{\infty} y(x)f(x)dx \tag{4}$$

$$\sigma^2 \equiv \text{variance of } x \equiv E[(x - <x>)^2] = \int_{-\infty}^{\infty} (x - <x>)^2 f(x)dx \tag{5}$$

$$\sigma \equiv \text{standard deviation of } x \equiv \text{SQRT}(\sigma^2) = \text{SQRT}\left[\int_{-\infty}^{\infty} (x - <x>)^2 f(x)dx\right] \tag{6}$$

## Some Known Probability Density Functions for Continuous Variables

### (A) Uniform Probability Density Function

A uniform probability density function can be represented generally by

$$f(x) = \begin{cases} \dfrac{1}{b-a} & \text{for } a < x < b \\ 0 & \text{for } x < a \text{ and } x > b \end{cases} \tag{7}$$

where $a$ and $b$ are constants.

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The mean $\mu$ and variance $\sigma^2$ of the continuous variable $x$ with the uniform probability density function given in Eqn (7) are given by:

$$\mu = \int_{-\infty}^{\infty} x.f(x)dx = \int_a^b \frac{x}{b-a}dx = \frac{0.5}{b-a}\Big[x^2\Big]_a^b = \frac{0.5}{b-a}(b^2-a^2) = 0.5(b+a)$$

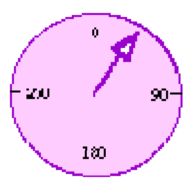
$$\sigma^2 = \int_{-\infty}^{\infty}(x-\mu)^2 f(x)dx = \int_a^b (x-\mu)^2 \frac{1}{b-a}dx = \int_a^b \frac{(x^2-2\mu x+\mu^2)}{b-a}dx$$

$$\sigma^2 = \frac{1}{b-a}\left[\frac{x^3}{3} - \mu x^2 + \mu^2 x\right]_a^b = \frac{1}{b-a}\left[x\left(\frac{x^2}{3} - \mu x + \mu^2\right)\right]_a^b$$

$$\sigma^2 = \frac{(b^2-a^2)}{3} - \mu(b-a) + \mu^2 = \frac{(b^2-a^2)}{3} - 0.5(b+a)(b-a) + 0.5^2(b+a)^2$$

$$\sigma^2 = (b+a)\left[\frac{(b-a)}{3} - \frac{(b-a)}{2} + \frac{(b+a)}{4}\right] = (b+a)\frac{(b+5a)}{12}$$

**Example 1** A spinning dial shown below will come to rest with the pointer aligned randomly with any of the degree markings in the range of 0–360 on the dial face. Determine the probability that the dial will come to rest with the pointer aligned in 20–65° range.




**Solution**
The dial pointer reading $R$ has a uniform probability density function given by

$$f(R) = \frac{1}{b-a} = \frac{1}{(360-0)^o} = \frac{1}{360^o}$$

$$P(20^o < R < 65^o) = \int_{20^o}^{65^o} f(R)dR = \int_{20^o}^{65^o}\frac{1}{360^o}dR = \frac{(65-20)^o}{360^o} = \frac{45}{360} = \frac{1}{8} = 0.125$$

**Example 2** The wait time $t$, in minutes, for a bus to arrive is uniformly distributed between 0 and 15 minutes. Determine (a) the probability density function of the wait time, (b) the average and standard deviation of the wait time, (c) the probability that a person would wait less than 12.5 minutes.

**Solution**
(a) If the probability density function of the wait time $t$ is uniformly distributed, it would be

given by $f(t) = \begin{cases} c & \text{for } 0\text{ min} < t < 15\text{ min} \\ 0 & \text{for } t > 15\text{ min} \end{cases}$  (2-1)

where $c$ is a constant.

But $1=\int_{-\infty}^{\infty} f(t)dt=\int_{0}^{15\min} c\,dt=c\Big[t\Big]_{0}^{15\min}=c15\min \qquad \therefore c=\dfrac{1}{15\min}$

Therefore, the probability density function of the wait time $t$ is given by

$$f(t)=\begin{cases}\dfrac{1}{15\min} & \text{for } 0\text{ min}<t<15\min\\ 0 & \text{for } t>15\min\end{cases}$$

(b) The average wait time is given by $E(t)=\int_{-\infty}^{\infty} tf(t)dt=\int_{0}^{15\min}\dfrac{t}{15\text{ min}}dt=\dfrac{15^2}{30}\min=7.5\text{ min}$

$$\sigma^2=\int_{-\infty}^{\infty}(t-<t>)^2 f(t)dt=\int_{0}^{15\text{ min}}\frac{t^2-2*7.5\min t+7.5^2\min^2}{15\text{ min}}dt$$

$$\sigma^2=\frac{1}{15\text{ min}}\left[\frac{t^3}{3}-7.5\min t^2+7.5^2\min^2 t\right]_0^{15\text{ min}}=\left(\frac{15^2}{3}-7.5*15+7.5^2\right)\min^2$$

$\sigma^2=18.75\ \min^2 \qquad \therefore \sigma=\pm 4.330\min$

(c) $P(t<12.5\text{ min})=\int_{0}^{12.5\text{ min}}\dfrac{1}{15\text{ min}}dt=\dfrac{12.5}{15}=0.8333$

**(B) Exponential Probability Density Function**

The exponential probability density function for a random continuous variable $x$ is given by

$$f(x)=\begin{cases}\lambda\exp(-\lambda x) & \text{for } x\geq 0\\ 0 & \text{for } x<0\end{cases} \tag{8}$$

where λ is a positive constant known as the rate or decay parameter. The corresponding cumulative probability distribution function is given by

$$F(x)\equiv\int_{-\infty}^{x} f(x)dx=\begin{cases}1-\exp(-\lambda x) & \text{for } x\geq 0\\ 0 & \text{for } x<0\end{cases} \tag{9}$$

The mean (or expected) value and variance of an exponentially distributed random variable $X$ with the rate parameter $\lambda$ are given, respectively, by

$$E(x)=\int_{-\infty}^{\infty} xf(x)dx=\int_0^{\infty}\underbrace{x}_{u}\underbrace{\lambda\exp(-\lambda x)dx}_{dv}\overset{\text{product rule}}{=}\left[\underbrace{x}_{u}\underbrace{\left(-\exp^{-\lambda x}\right)}_{v}\right]_0^{\infty}-\int_0^{\infty}\underbrace{\left(-\exp^{-\lambda x}\right)}_{v}\underbrace{dx}_{du}$$

$$E(x)=\left[-x\exp(-\lambda x)-\frac{1}{\lambda}\exp(-\lambda x)\right]_0^{\infty}=-\left[(x-\frac{1}{\lambda})\exp(-\lambda x)\right]_0^{\infty}\quad\text{or}$$

$$E(x)=-\left[0-\frac{1}{\lambda}*1\right]=\frac{1}{\lambda}\qquad(10)$$

$$\sigma^2=\int_0^{\infty}\underbrace{(x-\frac{1}{\lambda})^2}_{u}\underbrace{\lambda\exp(-\lambda x)dx}_{dv}\overset{\text{product rule}}{=}\left[\underbrace{(x-\frac{1}{\lambda})^2}_{u}\underbrace{\left(-\exp^{-\lambda x}\right)}_{v}\right]_0^{\infty}-\int_0^{\infty}\underbrace{\left(-\exp^{-\lambda x}\right)}_{v}\underbrace{2(x-\frac{1}{\lambda})dx}_{du}$$

$$\sigma^2=\left[\underbrace{(x-\frac{1}{\lambda})^2}_{u}\underbrace{\left(-\exp^{-\lambda x}\right)}_{v}\right]_0^{\infty}+\int_0^{\infty}2x\exp(-\lambda x)dx-\frac{2}{\lambda}\int_0^{\infty}\exp(-\lambda x)dx\qquad(11)$$

But $\int_0^{\infty}2x\exp(-\lambda x)dx=\frac{2}{\lambda}\int_0^{\infty}x\lambda\exp(-\lambda x)dx=\frac{2}{\lambda}E(x)=\frac{2}{\lambda}\left(\frac{1}{\lambda}\right)=\frac{2}{\lambda^2}$ and

$$\frac{2}{\lambda}\int\exp(-\lambda x)dx=-\frac{2}{\lambda^2}\exp(-\lambda x)$$

$$\therefore\sigma^2=\left[(x-\frac{1}{\lambda})^2\left(-\exp^{-\lambda x}\right)\right]_0^{\infty}+\frac{2}{\lambda^2}+\left[\frac{2}{\lambda^2}\exp(-\lambda x)\right]_0^{\infty}$$

$$\therefore\sigma^2=\left[0-\frac{1}{\lambda^2}(-1)\right]+\frac{2}{\lambda^2}+(0-\frac{2}{\lambda^2})=\frac{1}{\lambda^2}\qquad(12)$$

The exponential probability distribution function is used extensively used in the service, consumer product, and travel industries in connection with customer queue time analysis, reliability analysis, and traffic accident analysis. It is often used to determine the time until some specific event occurs. For example, the projected service life of a device (like battery), the time required for a fractional decay of a radio-active substance, the spending on a supermarket trip may all be estimated using the exponential probability distribution function.

**Example 3** Plutonium 239 decays continuously at a rate of 0.00284% per year. It follows that if $X$ is the time (in year) at which a randomly chosen plutonium atom will decay, the normalized probability density function for the plutonium atom decaying in the time $X$ will be given by $f(X) = a * \exp[-\frac{0.0000284}{\text{yr}} X]$, where $a$ is a constant to be determined. (a) Show that $a = \frac{0.0000284}{\text{yr}}$, and (b) determine the probability that a plutonium atom will decay between 100 and 500 years from now.

**Solution**

(a) For a normalized probability density function,

$$1.0 = \int_{-\infty}^{\infty} f(X)dX = \int_{0}^{\infty} a\exp\left(\frac{-0.0000284}{\text{yr}} X\right)dX = \frac{\text{yr}}{-0.0000284} a\exp\left(\frac{-0.0000284}{\text{yr}} X\right)\Bigg]_0^{\infty}$$

$$\text{or } 1.0 = \frac{\text{yr}}{-0.0000284} a(0-1) = \frac{\text{yr}}{0.0000284} a \text{ or } a = \frac{0.0000284}{\text{yr}}$$

(b) $P(100 \text{ yr} < X < 500 \text{ yr}) = \int_{100 \text{ yr}}^{500 \text{ yr}} f(X)dX = \int_{100 \text{ yr}}^{500 \text{ yr}} \frac{0.0000284}{\text{yr}} \exp\left(-\frac{0.0000284}{\text{yr}} X\right)dX$

$$P(100 \text{ yr} < X < 500 \text{ yr}) = -\exp\left(-\frac{0.0000284}{\text{yr}} X\right)\Bigg|_{100 \text{ yr}}^{500 \text{ yr}}$$

$$= -\exp(-0.0000284 * 500) + \exp(-0.0000284 * 100)$$

$$P(100 \text{ yr} < \text{X} < 500 \text{ yr}) = -0.9859 + 0.9972 = 0.0113$$

Thus, there is a 1.13% chance a plutonium atom will decay between 100 and 500 years from now.

**Example 4** The amount of time $t$ a postal clerk spends with a customer is known to be exponentially distributed with a mean $t$ value of 4.0 minutes. Determine (a) the exponential probability density function, and (b) the probability that a clerk will spend four to five minutes with a randomly selected customer.

**Solution**

(a) Recall that for the exponential probability density function of a continuous random variable $t$, $f(t) = \lambda \exp(-\lambda t)$, the expectation value of $t$ is given by Eqn (10) or $E(t) = \frac{1}{\lambda}$. For a mean $t$

value of 4.0 minutes, $E(t) = \frac{1}{\lambda} = 4.0 \text{ min}$. Therefore, $\lambda = \frac{1}{4 \text{ min}}$. It follows that the exponential probability density function is given by $f(t) = \frac{1}{4 \text{ min}} \exp(\frac{-t}{4 \text{ min}})$

(b) $P(4 \text{ min} < t < 5 \text{ min}) = \int_{4 \text{ min}}^{5 \text{ min}} \frac{1}{4 \text{ min}} \exp(\frac{-t}{4 \text{ min}}) dt = \left[ -\exp(\frac{-t}{4 \text{ min}}) \right]_{4 \text{ min}}^{5 \text{ min}}$

$P(4 \text{ min} < t < 5 \text{ min}) = -\left[ \exp(\frac{-5 \text{ min}}{4 \text{ min}}) - \exp(\frac{-4 \text{ min}}{4 \text{ min}}) \right] = -(e^{-1.25} - e^{-1}) = 0.0814$

**Example 5**

The service life $t$ of a computer part is exponentially distributed with a mean value of 10 years. Determine (a) the probability that the computer part lasts more than 7 years, and (b) the 80th percentile of the service life of the computer part (or the maximum service life of 80% of the computer part).

**Solution**

For the exponential probability density function of service life $t$, $f(t) = \lambda \exp(-\lambda t)$, the expectation value of $t$ is given by Eqn (10) or $E(t) = \frac{1}{\lambda}$. For a mean $t$ value of 10 years, $E(t) = \frac{1}{\lambda} = 10 \text{ yr}$. Therefore, $\lambda = \frac{1}{10 \text{ yr}}$. It follows that the exponential probability density function is given by $f(t) = \frac{1}{10 \text{ yr}} \exp(\frac{-t}{10 \text{ yr}})$.

(a) $P(t > 7 \text{ yr}) = 1 - P(t < 7 \text{ yr}) = 1 - \int_0^{7 \text{ yr}} \frac{0.1}{\text{yr}} \exp(\frac{-0.1t}{\text{yr}}) dt = 1 - \left[ -\exp(\frac{-0.1t}{\text{yr}}) \right]_0^{7 \text{ yr}}$

$P(t > 7 \text{ yr}) = 1 - P(t < 7 \text{ yr}) = 1 + (e^{-0.7} - 1) = 0.4966$

(b) Let the 80th percentile of the service life of the computer part be denoted by $t_{0.80}$. The 80th percentile of the service life of the computer part is given by

$$P(t < t_{0.80}) = 0.80 = \int_0^{t_{0.80}} \frac{1}{10 \text{ yr}} \exp(\frac{-t}{10 \text{ yr}}) dt = \left[ -\exp(\frac{-t}{10 \text{ yr}}) \right]_0^{t_{0.80}} = -\left[ \exp(\frac{-t_{0.80}}{10 \text{ yr}}) - 1 \right]$$

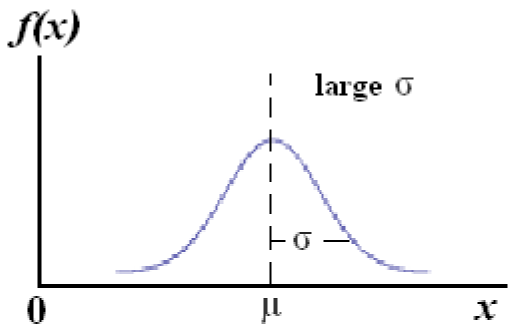
$\exp(\frac{-t_{0.80}}{10 \text{ yr}}) = 1 - 0.80 = 0.20$ or $\frac{-t_{0.80}}{10 \text{ yr}} = \ln(0.20) = -1.6094$ $\quad \therefore t_{0.80} = 16.094 \text{ yr}$

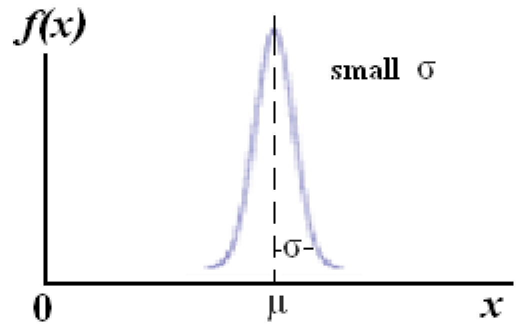
**(C) Normal (or Gaussian) Probability Density Function**

The normal (or Gaussian) probability density function is a two-parameter function given by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[\frac{-1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right] \tag{13}$$

where μ is the mean value for the continuous random variable $x$ and $\sigma$ is the standard deviation for $x$. As shown below, the function is represented by the well-known, symmetrical bell-shaped curve centered at $x = \mu$. The spread of the curve is determined by σ.



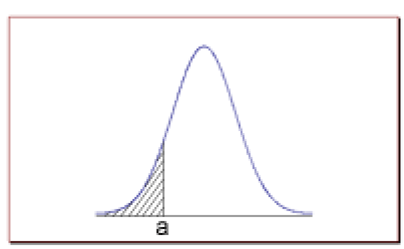

The normal distribution function has the following properties:

$$\int_{-\infty}^{\infty} f(x)dx = \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} \exp\left[\frac{-1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right]dx = 1.0 \tag{14}$$

$$E(x) = \int_{-\infty}^{\infty} xf(x)dx = \int_{-\infty}^{\infty} x\frac{1}{\sigma\sqrt{2\pi}} \exp\left[\frac{-1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right]dx = \mu \tag{15}$$

$$E[(x-\mu)^2] = \int_{-\infty}^{\infty} (x-\mu)^2 f(x)dx = \int_{-\infty}^{\infty} (x-\mu)^2 \frac{1}{\sigma\sqrt{2\pi}} \exp\left[\frac{-1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right]dx = \sigma^2 \tag{16}$$

- The probability that $x$ is less than $a$ is equal to the area under the normal curve bounded by $a$ and minus infinity (as indicated by the *shaded* area in the figure below).



- The probability that $x$ is greater than $a$ is equal to the area under the normal curve bounded by $a$ and plus infinity (as indicated by the *non-shaded* area in the figure above).
- About 68% of the area under the curve falls within 1 standard deviation of the mean.
- About 95% of the area under the curve falls within 2 standard deviations of the mean.
- About 99.7% of the area under the curve falls within 3 standard deviations of the mean.

The normal distribution function (particularly its standard counterpart, the standard normal distribution function--see the next section) is the most widely known and used of all probability density functions for continuous variables. It, for example, describes the distribution patterns found in many natural and engineering phenomena. The height, blood pressure, and IQ of a population are normally distributed, as well as measurement errors, and length of an object produced by a machine.

**Example 6** The service life $X$ (in days) of a light bulb manufactured by a company is a normal variate with a mean μ of 300 days and a standard deviation σ of 50 days. Determine the probability that a randomly-selected light bulb made by the company will have a service life of (a) at most 365 days, (b) at least 365 days.

**Solution**

(a) $P(X < 365 \text{ days}) = \int_0^{365} f(X)dX = \int_0^{365} \frac{1}{\sigma\sqrt{2\pi}} \exp\left[\frac{-1}{2}\left(\frac{X-\mu}{\sigma}\right)^2\right] dX$

Using the VBA numerical integration program (NumInteg.xls), $P(X < 365 \text{ days}) = 0.9032$

(b) $P(X > 365 \text{ days}) = \int_{365}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} \exp\left[\frac{-1}{2}\left(\frac{X-\mu}{\sigma}\right)^2\right] dX = 1 - \int_0^{365} \frac{1}{\sigma\sqrt{2\pi}} \exp\left[\frac{-1}{2}\left(\frac{X-\mu}{\sigma}\right)^2\right] dX$

$P(X > 365 \text{ days}) = 1 - P(X > 365 \text{ days}) = 1 - 0.9032 = 0.0968$

**Example 7** The scores on an IQ test administered to a large group of students are normally distributed with a mean μ of 100 and a standard deviation σ of 10. Determine the probability that a student will have an IQ score between 90 and 110.

**Solution**

$P(90 < X < 110) = \int_{90}^{110} f(X)dX = \int_{90}^{110} \frac{1}{\sigma\sqrt{2\pi}} \exp\left[\frac{-1}{2}\left(\frac{X-\mu}{\sigma}\right)^2\right] dX$

Using the VBA numerical integration program (NumInteg.xls), $P(90 < X < 110) = 0.6827$

**Example 8** The percentage of impurities in a product is a normally distributed variable $X$ with a mean $\mu_X = 1.8\%$ and a standard deviation $\sigma_X = 0.40\%$. For a randomly selected product, determine (a) the 95% confidence interval for $X$, (b) the probability that $X$ will fall outside the $(1.8 \pm 0.1)\%$ interval, (c) the probability that $X$ will exceed 2.0%, and (d) the probability that $X$ will fall within the 1:00-2:00% range.

**Solution**

(a) The 95% confidence interval for $X$ is defined by

$$X = \mu_X \pm c_{0.95} \text{ such that } P(\mu_X - c_{0.95} < X < \mu_X + c_{0.95}) = 0.95 \quad \text{or}$$

$$\int_{\mu_X - c_{0.95}}^{\mu_X + c_{0.95}} \frac{1}{\sigma_X \sqrt{2\pi}} \exp\left[\frac{-1}{2}\left(\frac{X-\mu_X}{\sigma_X}\right)^2\right] dX = \int_{1.8\% - c_{0.95}}^{1.8\% + c_{0.95}} \frac{1}{0.40\%\sqrt{2\pi}} \exp\left[\frac{-1}{2}\left(\frac{X-1.8\%}{0.40\%}\right)^2\right] dX = 0.95$$

Using the VBA numerical integration program (NumInteg.xls) in conjunction with a trial and error on $c_{0.95}$, $c_{0.95} = 0.7841$. Therefore, the 95% confidence interval for $X$ is given by

$$X = \mu_X \pm c_{0.95} = (1.8 \pm 0.7841)\% \quad \text{or} \quad 1.016\% < X < 2.584\%$$

(b) The probability that $X$ will fall outside the $(1.8 \pm 0.1)\%$ interval is given by

$$P(X < 1.7\,\% \text{ or } X > 1.9\,\%) = 1 - P(1.7\,\% < X < 1.9\,\%)$$

$$P(1.7\% < X < 1.9\%) = \int_{1.7\%}^{1.9\%} \frac{1}{0.40\%\sqrt{2\pi}} \exp\left[\frac{-1}{2}\left(\frac{X-1.8\%}{0.40}\right)^2\right] dX \overset{\substack{\text{using}\\ \textit{NumInteg}\\ \text{program}}}{=} 0.1974$$

Therefore, the probability that $X$ will fall outside the $(1.8 \pm 0.1)\%$ interval $= 1 - 0.1974$
$= 0.8026$

(c) The probability that $X$ will exceed 2.0 % is given by

$$P(X > 2.0\%) = 1 - P(X < 2.0\%) = 1 - \int_{0\%}^{2\%} \frac{1}{0.40\%\sqrt{2\pi}} \exp\left[\frac{-1}{2}\left(\frac{X-1.8\%}{0.40\%}\right)^2\right] dX$$

Using the VBA numerical integration program (*NumInteg*.xls),

$$P(X > 2\%) = 1 - 0.6914 = 0.3086$$

(d) The probability that $X$ will fall within the 1:00-2:00% range is given by

$$P(1.0\% < X < 2.0\%) = \int_{1\%}^{2\%} \frac{1}{0.40\%\sqrt{2\pi}} \exp\left[\frac{-1}{2}\left(\frac{X-1.8\%}{0.40\%}\right)^2\right] dX \overset{\substack{\text{using the}\\ \textit{NumInteg}\\ \text{program}}}{=} 0.6687$$

**(D) *Standard* Normal Probability Density Function or the Z-Distribution Function**

If one does not have the benefit of a calculator program such as *NumInteg.xls* that can numerically evaluate the cumulative probability, $P(X)=\int_0^X \frac{1}{\sigma\sqrt{2\pi}}\exp\left[\frac{-1}{2}\left(\frac{X-\mu}{\sigma}\right)^2\right]dX$,

one would have to refer to tables of cumulative normal probability in performing statistical analyses. Because the normal distribution function has two parameters, $\mu$ and $\sigma$, there is theoretically an infinite number of different possible combinations of $\mu$ and $\sigma$, and it is impractical to list the tables of cumulative probability values for all the different combinations of $\mu$ and $\sigma$ in a handbook of mathematics or statistics. Fortunately, the problem is resolved by an ingenious definition of the $Z$-variable, $Z(X)\equiv\frac{X-\mu_X}{\sigma_X}$ (17)

With this definition or transform, all the different distribution curves collapse into a single universal curve, all the different cumulative distribution tables consolidate into a single table and Eqn (13) simplifies to $f[Z(X)]=\frac{1}{\sqrt{2\pi}}\exp\left[\frac{-Z(X)^2}{2}\right]$ (18)

Eqn (18) defines the *standard* normal probability density function or simply the *Z*-distribution function. The *Z*-distribution function has the following properties:

$$\int_{-\infty}^{\infty} f(Z)dZ=\int_{-\infty}^{\infty}\frac{1}{\sqrt{2\pi}}\exp\left[\frac{-Z^2}{2}\right]dZ=1.0 \qquad (19)$$

$$E(Z)\equiv\mu_Z=\int_{-\infty}^{\infty} Z\frac{1}{\sqrt{2\pi}}\exp\left[\frac{-Z^2}{2}\right]dZ=0 \qquad (20)$$

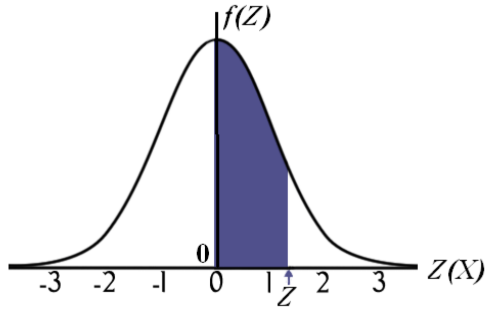
$$E[(Z-\mu_Z)^2]\equiv\sigma_Z{}^2=\int_{-\infty}^{\infty} Z^2\frac{1}{\sqrt{2\pi}}\exp\left[\frac{-Z^2}{2}\right]dZ=1.0 \text{ or } \sigma_Z=\pm1.0 \qquad (21)$$

The ***standard* normal distribution** is thus a special case of the normal distribution with a mean of zero and a standard deviation of one. The function is symmetrical about the mean *Z* of zero. It follows that the function corresponding to negative Z-values is a mirror image of the function corresponding to positive Z-values, viz., *f(−Z) = f(Z)*, and *P(−Z) = P(Z)*.

Table 1 presents the Z-table which lists the cumulative probability as a function of the Z-score. The cumulative probability corresponds to the area under the $f(Z)=\frac{1}{\sqrt{2\pi}}\exp\left[\frac{-Z^2}{2}\right]$ vs $Z$-curve between $Z=0$ & $Z=Z$. Thus, $P(Z=1.96)=\int_0^{1.96}\frac{1}{\sqrt{2\pi}}e^{\frac{-Z^2}{2}}dZ$ = Area shaded blue $\overset{\text{Table 1}}{=}$ 0.4750.

Table 1. *Z*-table listing the value of cumulative probability as a function of the *Z*-score.

$$P[Z(X)] = \int_0^{Z(X)} \frac{1}{\sqrt{2\pi}} \exp\left[\frac{-Z^2}{2}\right] dZ = \text{Area under the } f(Z) \text{ vs } Z \text{ curve between } Z = 0 \text{ and } Z = Z.$$




| Z | 0.00 | 0.01 | 0.02 | 0.03 | 0.04 | 0.05 | 0.06 | 0.07 | 0.08 | 0.09 |
|---|---|---|---|---|---|---|---|---|---|---|
| **0.0** | 0.0000 | 0.0040 | 0.0080 | 0.0120 | 0.0160 | 0.0199 | 0.0239 | 0.0279 | 0.0319 | 0.0359 |
| **0.1** | 0.0398 | 0.0438 | 0.0478 | 0.0517 | 0.0557 | 0.0596 | 0.0636 | 0.0675 | 0.0714 | 0.0753 |
| **0.2** | 0.0793 | 0.0832 | 0.0871 | 0.0910 | 0.0948 | 0.0987 | 0.1026 | 0.1064 | 0.1103 | 0.1141 |
| **0.3** | 0.1179 | 0.1217 | 0.1255 | 0.1293 | 0.1331 | 0.1368 | 0.1406 | 0.1443 | 0.1480 | 0.1517 |
| **0.4** | 0.1554 | 0.1591 | 0.1628 | 0.1664 | 0.1700 | 0.1736 | 0.1772 | 0.1808 | 0.1844 | 0.1879 |
| **0.5** | 0.1915 | 0.1950 | 0.1985 | 0.2019 | 0.2054 | 0.2088 | 0.2123 | 0.2157 | 0.2190 | 0.2224 |
| **0.6** | 0.2257 | 0.2291 | 0.2324 | 0.2357 | 0.2389 | 0.2422 | 0.2454 | 0.2486 | 0.2517 | 0.2549 |
| **0.7** | 0.2580 | 0.2611 | 0.2642 | 0.2673 | 0.2704 | 0.2734 | 0.2764 | 0.2794 | 0.2823 | 0.2852 |
| **0.8** | 0.2881 | 0.2910 | 0.2939 | 0.2967 | 0.2995 | 0.3023 | 0.3051 | 0.3078 | 0.3106 | 0.3133 |
| **0.9** | 0.3159 | 0.3186 | 0.3212 | 0.3238 | 0.3264 | 0.3289 | 0.3315 | 0.3340 | 0.3365 | 0.3389 |
| **1.0** | 0.3413 | 0.3438 | 0.3461 | 0.3485 | 0.3508 | 0.3531 | 0.3554 | 0.3577 | 0.3599 | 0.3621 |
| **1.1** | 0.3643 | 0.3665 | 0.3686 | 0.3708 | 0.3729 | 0.3749 | 0.3770 | 0.3790 | 0.3810 | 0.3830 |
| **1.2** | 0.3849 | 0.3869 | 0.3888 | 0.3907 | 0.3925 | 0.3944 | 0.3962 | 0.3980 | 0.3997 | 0.4015 |
| **1.3** | 0.4032 | 0.4049 | 0.4066 | 0.4082 | 0.4099 | 0.4115 | 0.4131 | 0.4147 | 0.4162 | 0.4177 |
| **1.4** | 0.4192 | 0.4207 | 0.4222 | 0.4236 | 0.4251 | 0.4265 | 0.4279 | 0.4292 | 0.4306 | 0.4319 |
| **1.5** | 0.4332 | 0.4345 | 0.4357 | 0.4370 | 0.4382 | 0.4394 | 0.4406 | 0.4418 | 0.4429 | 0.4441 |
| **1.6** | 0.4452 | 0.4463 | 0.4474 | 0.4484 | 0.4495 | 0.4505 | 0.4515 | 0.4525 | 0.4535 | 0.4545 |
| **1.7** | 0.4554 | 0.4564 | 0.4573 | 0.4582 | 0.4591 | 0.4599 | 0.4608 | 0.4616 | 0.4625 | 0.4633 |
| **1.8** | 0.4641 | 0.4649 | 0.4656 | 0.4664 | 0.4671 | 0.4678 | 0.4686 | 0.4693 | 0.4699 | 0.4706 |
| **1.9** | 0.4713 | 0.4719 | 0.4726 | 0.4732 | 0.4738 | 0.4744 | 0.4750 | 0.4756 | 0.4761 | 0.4767 |
| **2.0** | 0.4772 | 0.4778 | 0.4783 | 0.4788 | 0.4793 | 0.4798 | 0.4803 | 0.4808 | 0.4812 | 0.4817 |
| **2.1** | 0.4821 | 0.4826 | 0.4830 | 0.4834 | 0.4838 | 0.4842 | 0.4846 | 0.4850 | 0.4854 | 0.4857 |
| **2.2** | 0.4861 | 0.4864 | 0.4868 | 0.4871 | 0.4875 | 0.4878 | 0.4881 | 0.4884 | 0.4887 | 0.4890 |
| **2.3** | 0.4893 | 0.4896 | 0.4898 | 0.4901 | 0.4904 | 0.4906 | 0.4909 | 0.4911 | 0.4913 | 0.4916 |
| **2.4** | 0.4918 | 0.4920 | 0.4922 | 0.4925 | 0.4927 | 0.4929 | 0.4931 | 0.4932 | 0.4934 | 0.4936 |
| **2.5** | 0.4938 | 0.4940 | 0.4941 | 0.4943 | 0.4945 | 0.4946 | 0.4948 | 0.4949 | 0.4951 | 0.4952 |
| **2.6** | 0.4953 | 0.4955 | 0.4956 | 0.4957 | 0.4959 | 0.4960 | 0.4961 | 0.4962 | 0.4963 | 0.4964 |
| **2.7** | 0.4965 | 0.4966 | 0.4967 | 0.4968 | 0.4969 | 0.4970 | 0.4971 | 0.4972 | 0.4973 | 0.4974 |
| **2.8** | 0.4974 | 0.4975 | 0.4976 | 0.4977 | 0.4977 | 0.4978 | 0.4979 | 0.4979 | 0.4980 | 0.4981 |
| **2.9** | 0.4981 | 0.4982 | 0.4982 | 0.4983 | 0.4984 | 0.4984 | 0.4985 | 0.4985 | 0.4986 | 0.4986 |
| **3.0** | 0.4987 | 0.4987 | 0.4987 | 0.4988 | 0.4988 | 0.4989 | 0.4989 | 0.4989 | 0.4990 | 0.4990 |
| **3.1** | 0.4990 | 0.4991 | 0.4991 | 0.4991 | 0.4992 | 0.4992 | 0.4992 | 0.4992 | 0.4993 | 0.4993 |
| **3.2** | 0.4993 | 0.4993 | 0.4994 | 0.4994 | 0.4994 | 0.4994 | 0.4994 | 0.4995 | 0.4995 | 0.4995 |
| **3.3** | 0.4995 | 0.4995 | 0.4995 | 0.4996 | 0.4996 | 0.4996 | 0.4996 | 0.4996 | 0.4996 | 0.4997 |
| **3.4** | 0.4997 | 0.4997 | 0.4997 | 0.4997 | 0.4997 | 0.4997 | 0.4997 | 0.4997 | 0.4997 | 0.4998 |

To make use of the Z-table in a statistical analysis of a normally-distributed variable $X$, it is necessary to relate $X$ to $Z$ by means of Eqn (17) and then work in terms of $Z$. The approach is illustrated by re-analyzing Examples 6, 7, and 8 using the Z-statistics approach.

**Example 6** The service life $X$ (in days) of a light bulb manufactured by a company is a normal variate with a mean μ of 300 days and a standard deviation σ of 50 days. Determine the probability that a randomly-selected light bulb made by the company will have a service life of (a) at most 365 days, (b) at least 365 days.

**Alternate Solution**

(a) For $X = 365$ days, $Z = \dfrac{X-\mu_X}{\sigma_X} = \dfrac{365-300}{50} = 1.3$

$$P(X < 365 \text{ days}) = P(-\infty < Z < 1.3) = \overbrace{P(-\infty < Z < 0)}^{0.50 \text{ by symmetry property}} + \overbrace{P(0 < Z < 1.3)}^{\text{Table 1}} = 0.50 + 0.4032 = 0.9032$$

$$\text{(b) } P(X > 365 \text{ days}) = P(1.30 < Z < \infty) = \overbrace{P(0 < Z < \infty)}^{0.50 \text{ by symmetry property}} - \overbrace{P(0 < Z < 1.30)}^{P(Z=1.30),\ \text{Table 1}} = 0.50 - 0.4032 = 0.0968$$

**Example 7** The scores on an IQ test administered to a large group of students are normally distributed with a mean μ of 100 and a standard deviation σ of 10. Determine the probability that a student will have an IQ score between 90 and 110.

**Alternate Solution**

For $X = 90$ and $X = 110$, $Z = \dfrac{90-100}{10} = -1$ and $Z = \dfrac{110-100}{10} = 1$

$$P(90 < X < 110) = P(-1 < Z < 1) = \overbrace{P(-1 < Z < 0)}^{P(Z=-1)=P(Z1),\ \text{symmetry}} + \overbrace{P(0 < Z < 1)}^{P(Z=1)} = 2\overbrace{P(Z = 1)}^{\text{Table 1}} = 2 * 0.3413 = 0.6826$$

**Example 8** The percentage of impurities in a product is a normally distributed variable $X$ with a mean $\mu_X = 1.8\%$ and a standard deviation $\sigma_X = 0.40\%$. For a randomly selected product, determine (a) the 95% confidence interval for $X$, (b) the probability that $X$ will fall outside the $(1.8 \pm 0.1)\%$ interval, (c) the probability that $X$ will exceed 2.0%, and (d) the probability that $X$ will fall within the 1:00-2:00% range.

**Alternate Solution**

(a) The 95% confidence interval for $X$ is defined by

$X = \mu_X \pm c_{0.95}$ such that $P(\mu_X - c_{0.95} < X < \mu_X + c_{0.95}) = 0.95$ or

$$Z = \mu_Z \pm \Delta z = 0 \pm \Delta z \text{ such that } 0.95 = P(-\Delta z < Z < \Delta z) = \overbrace{P(-\Delta z < Z < 0)}^{P(Z=-\Delta z)=P(Z=\Delta z),\text{ symmetry}} + \overbrace{P(0 < Z < \Delta z)}^{P(Z=\Delta z)}$$

$2P(Z = \Delta z) = 0.95$ or $P(Z = \Delta z) = 0.475$

From Table 1, $P(Z) = 0.475$ corresponds to $Z = 1.960$ or more precisely, $Z = \pm 1.960$

$Z = \pm 1.960,$ the corresponding $X$ values are given by $X = \mu_X + Z\sigma_X = 1.8\% \pm 1.96 * 0.4\%$

Therefore, the 95% confidence interval for X is given by

$X = 1.8\% \pm 0.784\%$ or $1.016\% < X < 2.584\%$

(b) For $X = (1.8 \pm 0.1)\%,\ Z = \dfrac{X - \mu_X}{\sigma_X} = \dfrac{\pm 0.1}{0.4} = \pm 0.25$

The probability that $X$ will fall outside the (1.8 ± 0.1)% interval is given by

$P(X < 1.7\ \%$ or $X > 1.9\ \%) = P(Z < -0.25$ or $Z > 0.25) = P(Z < -0.25) + P(Z > 0.25)$

$$P(X < 1.7\ \% \text{ or } X > 1.9\ \%) \overset{\text{symmetry property}}{=} 2P(Z > 0.25)$$

$$P(X < 1.7\ \% \text{ or } X > 1.9\ \%) = 2[0.50 - P(0 < Z < 0.25)] \overset{\text{Table 1}}{=} 2(0.50 - 0.0987) = 0.8026$$

(c) For $X = 2.0\%,\ Z = \dfrac{X - \mu_X}{\sigma_X} = \dfrac{2 - 1.8}{0.40} = \dfrac{1}{2}$

The probability that $X$ will exceed 2.0 % is given by

$$P(X > 2.0\%) = P(Z > \frac{1}{2}) = \overbrace{P(0 < Z < \infty)}^{0.50} - \overbrace{P(0 < Z < \frac{1}{2})}^{P(Z=0.5)} \overset{\text{Table 1}}{=} 0.50 - 0.1915 = 0.3085$$

(d) For $X = 1.0\%$ and $2.0\%,\ Z = \dfrac{X - \mu_X}{\sigma_X} = \dfrac{1 - 1.8}{0.40} = -2$ and $\dfrac{2.0 - 1.8}{0.40} = \dfrac{1}{2}$

The probability that $X$ will fall within the 1:00-2:00% range is given by

$P(1.0\% < X < 2.0\%) = P(-2.0 < Z < 0.50) = P(-2.0 < Z < 0) + P(0 < Z < 0.50)$

$$P(1.0\% < X < 2.0\%) \overset{\substack{\text{symmetry}\\ \text{property}}}{=} P(0 < Z < 2.0) + P(0 < Z < 0.50) = P(Z = 2.0) + P(Z = 0.50)$$

$$P(1.0\% < X < 2.0\%) \overset{\text{Table 1}}{=} 0.4772 + 0.1915 = 0.6687$$

## Central Limit Theorem

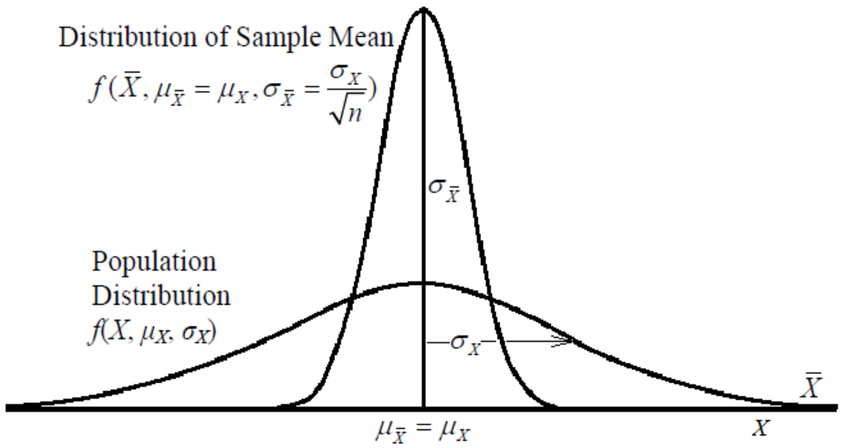
The Central Limit Theorem is used extensively in statistical analyses. It may be explained as follows. Given a distribution of a variable $X$–not necessarily a normal distribution–that has a mean $\mu_X$ and a variance $\sigma_X^2$, a second distribution, which is known as the distribution of the sample mean, may be created by repeatedly taking $n$ samples each time and determining the sample mean $\bar{X}$ each time. The Central Limit Theorem states that the distribution of the sample mean $\bar{X}$ has a mean $\mu_{\bar{X}}$, a variance $\sigma_{\bar{X}}^2$, and a standard deviation $\sigma_{\bar{X}}$ that are related to $\mu_X$, $\sigma_X^2$, and $\sigma_X$ of the original distribution as follows

$$\mu_{\bar{X}} = \mu_X \tag{22}$$

$$\sigma_{\bar{X}}^2 = \frac{\sigma_X^2}{n} \tag{23}$$

$$\sigma_{\bar{X}} = \frac{\sigma_X}{\sqrt{n}} \tag{24}$$

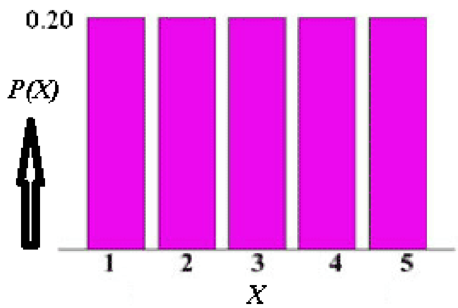
The figure below summarizes the relationship given in Eqn (22)−24. Notice that the distribution for the sample mean is narrower than the population distribution as a consequence of $\sigma_{\bar{X}} < \sigma_X$.




Moreover, the distribution of the sample mean $\bar{X}$ approaches a normal distribution as the sample size $n$ increases, ***even if the original distribution of X is other than a normal distribution***!

The Central Limit Theorem may be illustrated by the following computer simulation example based on a random sampling of a large population of $n$ distinct groups of balls present in equal proportions, i.e., balls in each group have a 1/5 or 20% chances of being picked. Balls in the five distinct groups are labeled 1, 2, 3, 4, and 5, respectively. Let $X$ denote the number label of a ball being picked.

For the uniform distribution of $X$,




Uniform (Non-Normal) Distribution of $X$

it may be shown that the mean $\mu_X$ and variance $\sigma_X^2$ of $X$ are given, respectively, by

$$\mu_X = \sum_{i=1}^{5} P_i X_i = \frac{1}{5}(1+2+3+4+5) = 3.0 \qquad (25)$$

$$\sigma_X{}^2 = \sum_{i=1}^{5} P_i (X_i - \mu_X)^2 = \frac{1}{5}(4+1+0+1+4) = 2.0 \qquad (26)$$

Suppose $n$ balls are randomly picked by the computer program for $N$ successive times, each time the average of the $n$ number labels is computed. The sample mean $\overline{X}$ and its variance $s_{\overline{X}}{}^2$ would be given, respectively, by

$$\overline{X} = \frac{1}{n}\sum_{i=1}^{5} X_i \qquad (27)$$

$$s_{\overline{X}}{}^2 = \frac{1}{n-1}\sum_{i=1}^{n} (X_i - \overline{X})^2 \qquad (28)$$

For sample size $n = 5$ and $N = 6$ successive averaging trials, the results shown in Figure 1 may be obtained. The red bar denotes the drawn label $X$ and the blue bar denotes the sample mean $\overline{X}$ in each trial. The distribution of the sample mean $\overline{X}$ in the $N = 6$ trials is given in Figure 2.

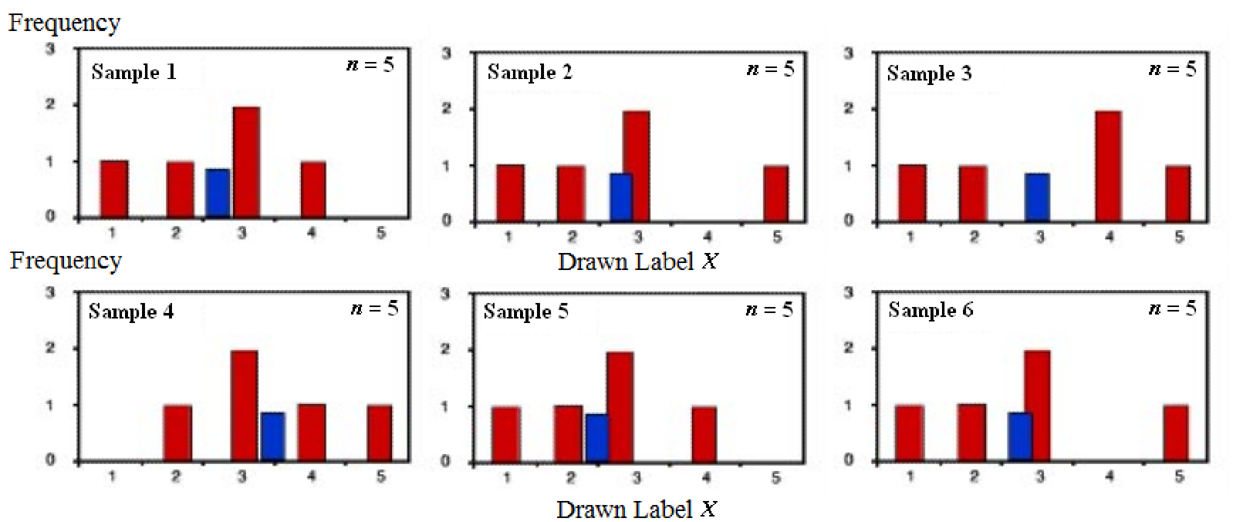



Figure 1. Frequency distribution of randomly-drawn label X for a sample size n = 5 and N = 6 successive trials.

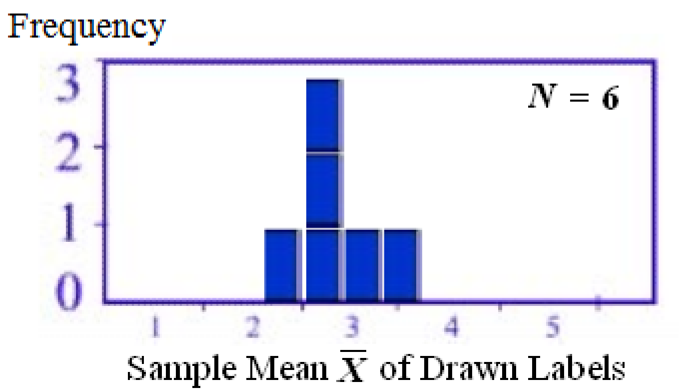



**Figure 2.** Distribution of sample mean $\bar{X}$ for $N$ = 6.

The sample means of the six samples are not the same (see Figure 1) and Figure 2 shows that the distribution of the sample means clusters in a narrow range that suggests a bell-shaped distribution characteristic of a normal distribution. As the number of averaging trials $N$ is increased from 6, the distribution of sample mean $\bar{X}$ becomes "smoother" and more closely resembles a normal distribution. For example, when the sample size is increased from 6 to 25, the distribution of sample mean $\bar{X}$ shown in Table 1 is obtained.

Table 1. Frequency distribution of sample mean $\bar{X}$ for $N = 25$ averaging trials.

| **Sample Mean** $\bar{X}$ | 1.40 | 2.60 | 2.80 | 3.00 | 3.20 | 3.40 | 3.60 | 3.80 | 4.00 | 4.20 |
|---|---|---|---|---|---|---|---|---|---|---|
| **Frequency** | 1 | 3 | 6 | 1 | 4 | 4 | 1 | 3 | 1 | 1 |

As shown in Figures 3 and 3', the distribution of the sample mean $\bar{X}$ becomes progressively more normal-like as the number $N$ of samples (or averaging trials) is increased. In the limit of $N \rightarrow \infty$ (or ~1000 for most practical purposes), the distribution becomes normal. Note that as $N \rightarrow \infty$, the mean value $\mu_{\bar{X}}$ and standard deviation $\sigma_{\bar{X}}$ of $\bar{X}$ are as predicted by Eqn (22) and (24).

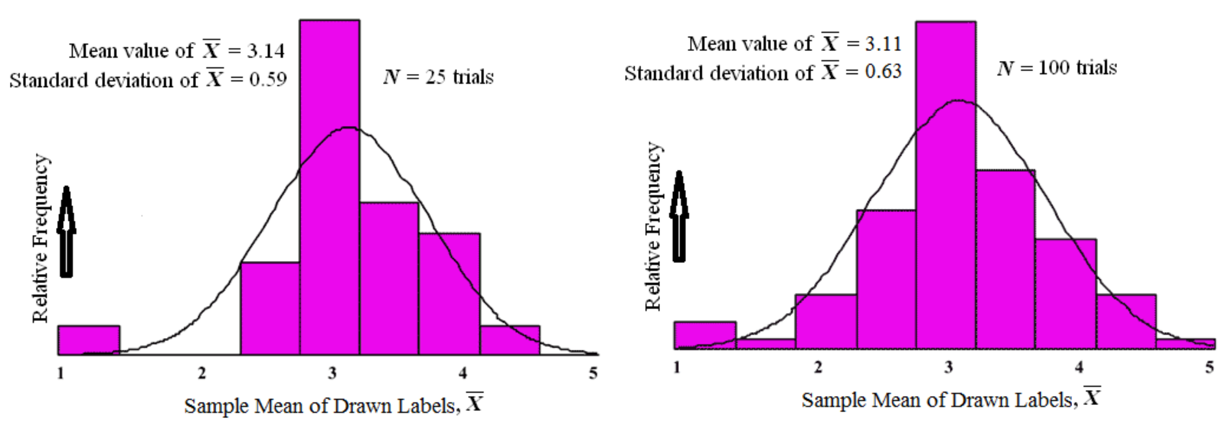



Figure 3. Progressive shift of distribution toward the normal distribution as the $N$ number of trials isincreased.

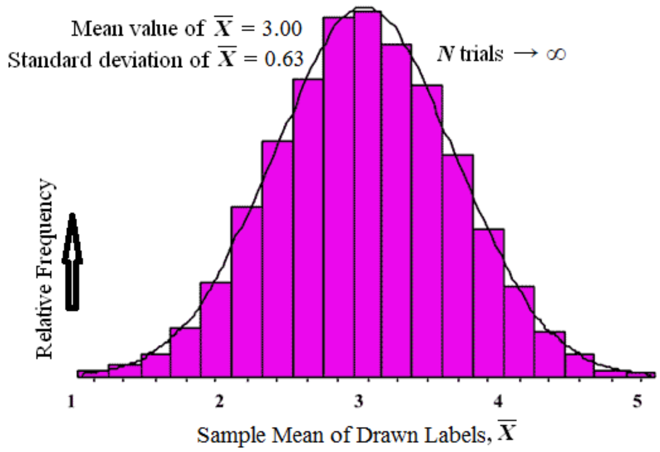



Figure 3'. Distribution becomes normal distribution when $N$ gets sufficiently large (~ 1000).

The foregoing computer simulation example illustrates an important result of general validity, namely that, the distribution of the sample mean $\overline{X}$ drawn from a population with ***any kind*** of distribution can be approximated by the normal distribution when the sample size $n$ is sufficiently large ($n \geq 30$) or when the number $N$ of averaging trials is sufficiently large ($N$ ~ 100–1000). The larger the sample size $n$ in each of the averaging trials or the larger the number of the averaging trials ($N$), the better is the approximation of the sample mean distribution to the normal distribution. Moreover, the mean value and standard deviation of the sample mean $\overline{X}$ are given by Eqn (22) and (24), respectively, where $n$ is the sample size in each averaging trial.

**Example 9** A grain company markets rice in 25-lb sacks. Data collected over many years indicate that the weight $W$ of rice shipped in each sack is a normally distributed variable with a mean value $\mu_W = 25.0$ lb and a standard deviation $\sigma_W = 0.25$ lb. Determine the probability that a sack will contain (a) less than 25.25 lb of rice, and (b) between 24.50 and 25.25 lb of rice.

If a delivery truck ships 9 sacks of rice, what is the probability that the *average* of the shipment contains less than 25.25 lb per sack?

**Solution**

Use the $Z$-distribution function and the definition of $Z$ to relate $Z$ to $W$, $Z = \dfrac{W - \mu_W}{\sigma_W}$.

(a) For $W = 25.25$ lb, $Z = \dfrac{25.25 - 25.0}{0.25} = 1.0$

$$P(W < 25.25 \text{ lb}) = P(Z < 1.0) = P(-\infty < Z < 0) + P(0 < Z < 1.0)$$

But symmetry property gives $P(-\infty < Z < 0) = P(0 < Z < \infty) = \frac{1}{2} = 0.50$ and $P(0 < Z < 1.0) = A(Z = 1.0) = 0.3413$.

Therefore,
$$\begin{aligned} P(W < 25.25 \text{ lb}) &= P(-\infty < Z < 0) + P(0 < Z < 1.0) \\ &= 0.50 + 0.3413 \\ &= 0.8413 \end{aligned}$$

(b) For $W = 25.25$ lb, $Z = 1.0$ from part a above.

For $W = 24.50$ lb, $Z = \dfrac{24.50 - 25.0}{0.25} = \dfrac{-0.50}{0.25} = -2.0$

$P(24.50 \text{ lb} < W < 25.25 \text{ lb}) = P(-2.0 < Z < 1.0)$

But $P(-2.0 < Z < 1.0) = P(-2.0 < Z < 0) + P(0 < Z < 1.0)$
$= P(0 < Z < 2.0) + P(0 < Z < 1.0)$
$= A(Z = 2.0) + A(Z = 1.0)$
$= 0.4772 + 0.3413$
$= 0.8185$

Therefore, $P(24.50 \text{ lb} < W < 25.25 \text{ lb}) = 0.8185$

(c) According to the Central Limit Theorem, the *average* of the normally distributed $W$ of 9 sacks, $\overline{W}$, is also a normally distributed variable with a mean $\mu_{\overline{W}}$ that is the same as $W$ ($\mu_W$) but with a standard $\sigma_{\overline{W}}$ that is different from that ($\sigma_W$) of $W$. Specifically,

Eqn (20) => $\mu_{\overline{W}} = \mu_W = 25.0$ lb

Eqn (22) => $\sigma_{\overline{W}} = \dfrac{\sigma_W}{\sqrt{n}} = \dfrac{0.25 \text{ lb}}{\sqrt{9}} = 0.0833 \text{ lb}$

For $\overline{W} = 25.25\text{lb}$, $Z = \dfrac{\overline{W} - \mu_{\overline{W}}}{\sigma_{\overline{W}}} = \dfrac{25.25 - 25.0}{0.0.0833} = \dfrac{0.25}{0.0833} = 3.0$

Therefore, $P(\overline{W} < 25.25 \text{ lb}) = P(Z < 3.0) = P(-\infty < Z < 0) + P(0 < Z < 3.0)$
$P(\overline{W} < 25.25 \text{ lb}) = 0.50 + A(Z{=}3.0)$
$= 0.50 + 0.4987$
$= 0.9987$

**Example 10** Based on data collected from the National Health Survey, the weight for adult males in the United States follows a normal distribution with a mean of 173 lb and a standard deviation of 30 lb. (a) If one U.S. adult male is randomly selected, what is the probability that his weight will be greater than 180 lb? (b) If 36 different U.S. adult males are randomly selected, what is the probability that their sample mean weight will be greater than 180 lb?

**Solution**

(a) Using the $Z$-statistics approach, relate $Z$ to $W$, $Z = \dfrac{W - \mu_W}{\sigma_W}$.

For $W = 180$ lb, $Z = \dfrac{180 - 173}{30} = 0.233$

$P(W > 180 \text{ lb}) = P(Z > 0.233) = 0.50 - P(Z < 0.233) = 0.50 - A(Z = 0.233)$

From Z-table, $A(Z = 0.233) = 0.0923$.
Therefore, $P(W > 180 \text{ lb}) = 0.50 - 0.0923 = 0.4077$

(b) Using the Central Limit theorem, Eqn (20) => $\mu_{\bar{X}} = \mu_X = 173$ lb

$$\text{Eqn (23)} \Rightarrow \sigma_{\bar{X}} = \frac{\sigma_X}{\sqrt{n}} = \frac{30 \text{ lb}}{\sqrt{36}} = 5 \text{ lb}$$

For $\bar{W}$ = 180 lb, $Z = \dfrac{\bar{W} - \mu_{\bar{W}}}{\sigma_{\bar{W}}} = \dfrac{180 - 173}{5} = 1.40$

$P(\bar{W} > 180 \text{ lb}) = P(Z > 1.40) = 0.50 - P(Z < 1.40) = 0.50 - A(Z = 1.40)$

From Z-table, $A(Z = 1.40) = 0.4192$.
Therefore, $P(\bar{W} > 180 \text{ lb}) = 0.50 - 0.4192 = 0.0808$

**Example 11** Measurements on 120 random trucks on a highway give a mean speed of 62 mph. Given that the truck population on the highway has a mean of 60 mph and a standard deviation of 12.5 mph, find the probability that the average speed of 120 trucks will exceed 62 mph.

## Solution

Using the Z-statistics approach, $Z = \dfrac{\bar{u} - \mu_{\bar{u}}}{\sigma_{\bar{u}}} = \dfrac{\bar{u} - \mu_u}{\sigma_u/\sqrt{n}} = \dfrac{62 - 60}{12.5/\sqrt{120}} = \dfrac{2}{1.141} = 1.753$

$$P(\bar{u} > 62 \text{ mph}) = P(Z > 1.753) = P(0 < Z < \infty) - P(Z < 1.753) = 0.50 - A(Z = 1.753)$$

From the Z-table, $A(Z = 1.753) = 0.4602$. Therefore, $P(\bar{u} > 62 \text{ mph}) = 0.50 - 0.4602 = 0.0382$

**Example 12** A sardine packaging company claims that its cans of sardine have a normal weight distribution with a mean of 6.05 oz and a standard deviation of 0.18 oz. An independent check by a consumer group found that a random case of 36 cans of sardine gives an average weight of 5.97 oz. What is the probability of the observation of the test result if the company's claim is valid, i.e., determine $P(X \le 5.97 \text{ oz})$. What conclusion may be drawn by the consumer group?

**Solution**

According to the Central Limit Theorem, $\mu_{\bar{X}} = \mu_X = 6.05$ oz and $\sigma_{\bar{X}} = \dfrac{\sigma_X}{\sqrt{n}} = \dfrac{0.18 \text{ oz}}{\sqrt{36}} = 0.03$ oz

Use the Z-statistics approach. $Z = \dfrac{\bar{X} - \mu_{\bar{X}}}{\sigma_{\bar{X}}} = \dfrac{5.97 - 6.05}{0.03} = -2.667$

$$P(X \le 5.97 \text{ oz}) = P(Z \le -2.667) \overset{\text{symmetry}}{=} P(Z \ge 2.667) = P(0 < Z < \infty) - P(Z \le 2.667)$$

$$P(X \le 5.97 \text{ oz}) = 0.50 - A(Z = 2.667) \overset{\text{Table 1}}{=} 0.5000 - 0.4962 = 0.0038$$

Since the calculated probability for the observation of the result is 0.38%, which is very low, the consumer group may be justified to conclude that the company's claim is false.

**(E) Student's t-Probability Density Function**

In order to use the normal or *Z*- distribution function in a statistical analysis, it is necessary that the standard deviation $\sigma_X$ of the population of the random continuous *X*-variable be known or the sample size *n* upon which the sample mean $\bar{X}$ is based is sufficiently large ($n > 30$). If neither of these two conditions is met, the use of normal or Z- statistic is not justified. One must instead use the Student's *t*-distribution function or, simply, the *t*-distribution, which has the probability density function defined by

$$f(t) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\,\Gamma\left(\frac{\nu}{2}\right)}\left(1+\frac{t^2}{\nu}\right)^{\frac{-1}{2}(\nu+1)}, \qquad (29)$$

where ν is the number of *degrees of freedom* and Γ is the *Gamma function*.

For ν even,
$$\frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\,\Gamma\left(\frac{\nu}{2}\right)} = \frac{(\nu-1)(\nu-3)\ldots 5*3}{2\sqrt{\nu}(\nu-2)(\nu-4)\ldots 4*2} \qquad (30)$$

For ν odd,
$$\frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\,\Gamma\left(\frac{\nu}{2}\right)} = \frac{(\nu-1)(\nu-3)\ldots 4*2}{2\sqrt{\nu}(\nu-2)(\nu-4)\ldots 5*3} \qquad (31)$$

Figures 4 and 4' present representative plots of the probability density function *f(t)* and the

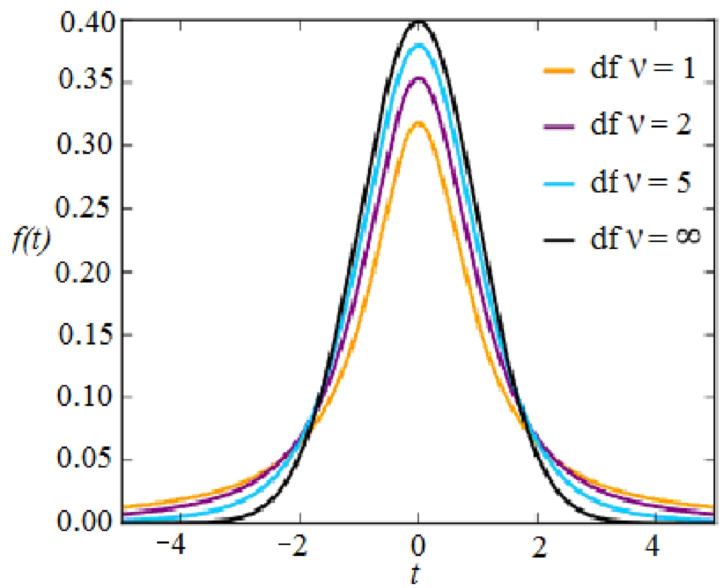



Figure 4. *t*-Distribution: Probability density *f(t)* as a function of continuous random variable *t*.

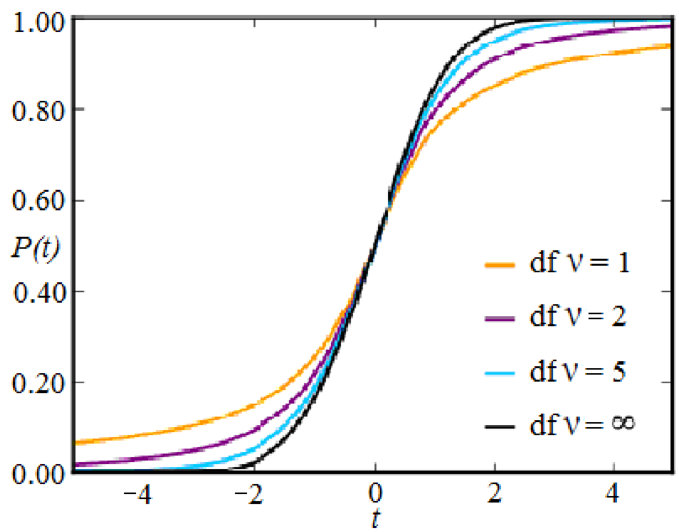



Figure 4'. $t$-Distribution: Cumulative probability $P(t)$ as a function of continuous random variable $t$.

cumulative probability $P(t)=\int_{-\infty}^{t} f(t)dt$ of $t$−distribution with the degree of freedom ν as a parameter. The degree of freedom ν is given by $\nu = N_{\text{data pts}} - N_{\text{indept variables}}$ (32)

where $N_{\text{data pts}} \equiv (\text{\# data points/measurements})$ and $N_{\text{indept variables}} \equiv (\text{\# independent variables})$.

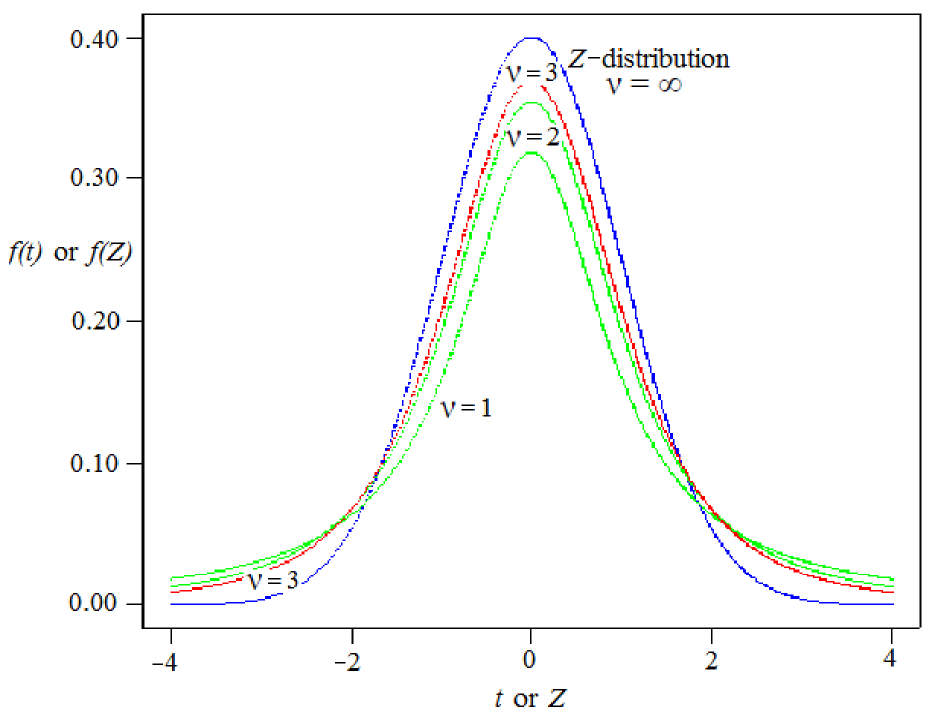



Figure 5. A comparison of $t$−distribution with $Z$−distribution.

The overall shape of the probability density function of the $t$-distribution resembles the bell shape of a normally distributed variable with mean 0 and variance 1, except that the shape of the $t$-distribution is flatter than that of the normal distribution, and the $t$-distribution has greater area under the tails. Figure 5 shows a comparison of the $t$-distribution with $Z$-distribution. The Z-scores defining a 95% confidence interval are ±1.960 and the tail areas above $Z = 1.960$ and below $Z = –1.960$ each has an area of 0.025. The confidence interval (or area under the curve) bounded by the $t = ±1.960$ scores is 92.76% (< 95%) for $v = 3$ and the tail areas above $t = 1.960$ and below $t = –1.960$ each has an area of 0.0362 (> 0.025). The fact that the $t$-distribution has higher tail areas than the $Z$-distribution means that critical $t$-values must be set higher than the corresponding critical $Z$-values in statistical analyses. As $v$ increases above 3, the $t$-distribution approaches the $Z$-distribution, and the difference in the tail areas corresponding to the same critical $t$- and $Z$-scores narrows progressively. Thus, for $v = 30$, the critical $t$-scores of ±1.960 give tail areas of 0.0297 (closer to 0.0250). For most practical purposes, the $t$-distribution may be substituted by the $Z$-distribution for $v \geq 30$.

The $t$-distribution has the following properties:

- the mean of the distribution ($\mu_t$) is equal to 0 and the distribution is centered at $t = 0$ and symmetric about $t = 0$, i.e., $f(–t) = f(t)$
- the variance is equal to $v / ( v - 2 )$, where $v$ is the degrees of freedom and $v \geq 2$
- the variance is always greater than 1, although it is close to 1 when there are many degrees of freedom
- the total area under the $f(t)$-vs-$t$ curve is 1.0 and each half of the curve above and below $t = 0$ has an area of ½.

The symbol $t_\alpha$ is used to represent the $t$-value for which the area to the right is α.

Table 2 presents the $t$-distribution table with the values of $t_{\nu,\alpha}$ as a function of the degree of freedom $v$ and one-sided or two-sided cumulative probability. The symbol $t_{\nu,\alpha}$ is used to denote the $t$-value for which the area (under $f(t)$-vs-$t$ curve) to the right is α for a specified degree of freedom $v$. Thus, $t_{\nu=10,\alpha=0.025} = 2.228$ means that the tail area to the right of $t = 2.228$ is 0.025 for $v = 10$. A $t$-distribution calculator is available on the web site, [http://stattrek.com/Tables/T.aspx](http://stattrek.com/Tables/T.aspx), which allows a determination of the value of the third parameter if any two of the parameters, ν, cumulative probability $P \equiv (1 – \alpha)$, and $t$-value are specified. Thus, if ν and $(1 – \alpha)$ are specified as $\nu = 10$ and $P = (1 – \alpha) = 0.975$, the calculator program will return a value of 2.228 for the $t$-value. Likewise, if ν and $t$-value are specified as $\nu = 10$ and $t$-value = 2.228, the program will determine $P = (1 – \alpha) = 0.975$ or $\alpha = 0.025$.

**Example 13** A survey of 31 students reveals that the average one-way travel to a college is 19.4 min with a standard deviation of 9.6 min. Determine the 95% confidence for the average one-way travel time to the college.

**Solution**

The 95% confidence interval for the average one-way travel time to the college centers about the

Table 2. $t$-Table: Values of $t_{v,\alpha}$ as a function of the degree of freedom $v$ and one-sided or two-sided cumulative probability, e.g., $t_{v=10,\alpha=0.025} = 2.228$.

| ***α*** | **0.30** | **0.25** | **0.20** | **0.15** | **0.10** | **0.05** | **0.025** | **0.010** | **0.005** | **0.0025** | **0.001** | **0.0005** |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ***One Sided*** | **70%** | **75%** | **80%** | **85%** | **90%** | **95%** | **97.5%** | **99%** | **99.5%** | **99.75%** | **99.9%** | **99.95%** |
| ***Two Sided*** | **35%** | **50%** | **60%** | **70%** | **80%** | **90%** | **95%** | **98%** | **99%** | **99.5%** | **99.8%** | **99.9%** |
| **1** | 0.727 | 1.000 | 1.376 | 1.963 | 3.078 | 6.314 | 12.71 | 31.82 | 63.66 | 127.3 | 318.3 | 636.6 |
| **2** | 0.617 | 0.816 | 1.061 | 1.386 | 1.886 | 2.920 | 4.303 | 6.965 | 9.925 | 14.09 | 22.33 | 31.60 |
| **3** | 0.584 | 0.765 | 0.978 | 1.250 | 1.638 | 2.353 | 3.182 | 4.541 | 5.841 | 7.453 | 10.21 | 12.92 |
| **4** | 0.569 | 0.741 | 0.941 | 1.190 | 1.533 | 2.132 | 2.776 | 3.747 | 4.604 | 5.598 | 7.173 | 8.610 |
| **5** | 0.559 | 0.727 | 0.920 | 1.156 | 1.476 | 2.015 | 2.571 | 3.365 | 4.032 | 4.773 | 5.893 | 6.869 |
| **6** | 0.553 | 0.718 | 0.906 | 1.134 | 1.440 | 1.943 | 2.447 | 3.143 | 3.707 | 4.317 | 5.208 | 5.959 |
| **7** | 0.549 | 0.711 | 0.896 | 1.119 | 1.415 | 1.895 | 2.365 | 2.998 | 3.499 | 4.029 | 4.785 | 5.408 |
| **8** | 0.546 | 0.706 | 0.889 | 1.108 | 1.397 | 1.860 | 2.306 | 2.896 | 3.355 | 3.833 | 4.501 | 5.041 |
| **9** | 0.543 | 0.703 | 0.883 | 1.100 | 1.383 | 1.833 | 2.262 | 2.821 | 3.250 | 3.690 | 4.297 | 4.781 |
| **10** | 0.542 | 0.700 | 0.879 | 1.093 | 1.372 | 1.812 | 2.228 | 2.764 | 3.169 | 3.581 | 4.144 | 4.587 |
| **11** | 0.540 | 0.697 | 0.876 | 1.088 | 1.363 | 1.796 | 2.201 | 2.718 | 3.106 | 3.497 | 4.025 | 4.437 |
| **12** | 0.539 | 0.695 | 0.873 | 1.083 | 1.356 | 1.782 | 2.179 | 2.681 | 3.055 | 3.428 | 3.930 | 4.318 |
| **13** | 0.538 | 0.694 | 0.870 | 1.079 | 1.350 | 1.771 | 2.160 | 2.650 | 3.012 | 3.372 | 3.852 | 4.221 |
| **14** | 0.537 | 0.692 | 0.868 | 1.076 | 1.345 | 1.761 | 2.145 | 2.624 | 2.977 | 3.326 | 3.787 | 4.140 |
| **15** | 0.536 | 0.691 | 0.866 | 1.074 | 1.341 | 1.753 | 2.131 | 2.602 | 2.947 | 3.286 | 3.733 | 4.073 |
| **16** | 0.535 | 0.690 | 0.865 | 1.071 | 1.337 | 1.746 | 2.120 | 2.583 | 2.921 | 3.252 | 3.686 | 4.015 |
| **17** | 0.534 | 0.689 | 0.863 | 1.069 | 1.333 | 1.740 | 2.110 | 2.567 | 2.898 | 3.222 | 3.646 | 3.965 |
| **18** | 0.534 | 0.688 | 0.862 | 1.067 | 1.330 | 1.734 | 2.101 | 2.552 | 2.878 | 3.197 | 3.610 | 3.922 |
| **19** | 0.533 | 0.688 | 0.861 | 1.066 | 1.328 | 1.729 | 2.093 | 2.539 | 2.861 | 3.174 | 3.579 | 3.883 |
| **20** | 0.533 | 0.687 | 0.860 | 1.064 | 1.325 | 1.725 | 2.086 | 2.528 | 2.845 | 3.153 | 3.552 | 3.850 |
| **21** | 0.532 | 0.686 | 0.859 | 1.063 | 1.323 | 1.721 | 2.080 | 2.518 | 2.831 | 3.135 | 3.527 | 3.819 |
| **22** | 0.532 | 0.686 | 0.858 | 1.061 | 1.321 | 1.717 | 2.074 | 2.508 | 2.819 | 3.119 | 3.505 | 3.792 |
| **23** | 0.532 | 0.685 | 0.858 | 1.060 | 1.319 | 1.714 | 2.069 | 2.500 | 2.807 | 3.104 | 3.485 | 3.767 |
| **24** | 0.531 | 0.685 | 0.857 | 1.059 | 1.318 | 1.711 | 2.064 | 2.492 | 2.797 | 3.091 | 3.467 | 3.745 |
| **25** | 0.531 | 0.684 | 0.856 | 1.058 | 1.316 | 1.708 | 2.060 | 2.485 | 2.787 | 3.078 | 3.450 | 3.725 |
| **26** | 0.531 | 0.684 | 0.856 | 1.058 | 1.315 | 1.706 | 2.056 | 2.479 | 2.779 | 3.067 | 3.435 | 3.707 |
| **27** | 0.531 | 0.684 | 0.855 | 1.057 | 1.314 | 1.703 | 2.052 | 2.473 | 2.771 | 3.057 | 3.421 | 3.690 |
| **28** | 0.530 | 0.683 | 0.855 | 1.056 | 1.313 | 1.701 | 2.048 | 2.467 | 2.763 | 3.047 | 3.408 | 3.674 |
| **29** | 0.530 | 0.683 | 0.854 | 1.055 | 1.311 | 1.699 | 2.045 | 2.462 | 2.756 | 3.038 | 3.396 | 3.659 |
| **30** | 0.530 | 0.683 | 0.854 | 1.055 | 1.310 | 1.697 | 2.042 | 2.457 | 2.750 | 3.030 | 3.385 | 3.646 |
| **40** | 0.529 | 0.681 | 0.851 | 1.050 | 1.303 | 1.684 | 2.021 | 2.423 | 2.704 | 2.971 | 3.307 | 3.551 |
| **50** | 0.528 | 0.679 | 0.849 | 1.047 | 1.299 | 1.676 | 2.009 | 2.403 | 2.678 | 2.937 | 3.261 | 3.496 |
| **60** | 0.527 | 0.679 | 0.848 | 1.045 | 1.296 | 1.671 | 2.000 | 2.390 | 2.660 | 2.915 | 3.232 | 3.460 |
| **80** | 0.526 | 0.678 | 0.846 | 1.043 | 1.292 | 1.664 | 1.990 | 2.374 | 2.639 | 2.887 | 3.195 | 3.416 |
| **100** | 0.526 | 0.677 | 0.845 | 1.042 | 1.290 | 1.660 | 1.984 | 2.364 | 2.626 | 2.871 | 3.174 | 3.390 |
| **120** | 0.526 | 0.677 | 0.845 | 1.041 | 1.289 | 1.658 | 1.980 | 2.358 | 2.617 | 2.860 | 3.160 | 3.373 |
| $\infty$ | 0.524 | 0.674 | 0.842 | 1.036 | 1.282 | 1.645 | 1.960 | 2.326 | 2.576 | 2.807 | 3.090 | 3.291 |

mean with two tails. Therefore, $\alpha \overset{\text{95% confidence interval}}{\overbrace{=}} \frac{1-0.95}{2} = 0.025$

$$t_{\nu,\alpha} = \frac{\bar{X} - <\bar{X}>}{s_{\bar{X}}} = \frac{\bar{X} - <X>}{s_X/\sqrt{31}} = \frac{\bar{X} - 19.4 \text{ min}}{9.6 \text{ min}/\sqrt{31}} = \frac{\bar{X} - 19.4 \text{ min}}{1.724 \text{ min}}$$

$\nu = 31 - 1 = 30, \qquad \alpha = 0.025$

From the *t*-table (Table 2), $t_{\nu=30,\alpha=0.025} = \pm 2.042$

$$\therefore \bar{X} = <\bar{X}> \pm t_{\nu,\alpha} s_{\bar{X}} = (19.4 \pm 2.042 * 1.724) \text{min} = (19.4 \pm 3.517) \text{min}$$

Therefore, the 95% confidence interval of the mean one-way travel time to the college is given by 15.883 min < $\bar{X}$ < 22.917 min

**Example 14** The saturated vapor pressure $P^{sat}$ of a newly synthesized compound has been measured five times at 40°C with the following results: 125.0, 124.2, 124.7, 126.0, and 125.3 mm Hg. Determine (a) the sample mean, $<P^{sat}>$, and its standard deviation, $s_P$, and (b) the 95% and 99% confidence intervals for $P^{sat}$.

**Solution**

(a) $<P^{sat}>$ = (125.0 + 124.2 + 124.7 + 126.0 + 125.3)/5 mm Hg = 125.04 mm Hg

$$\text{Variance } s_P^2 = \sum_{i=1}^{n=5} \frac{1}{n-1}(P^{sat}{}_i - <P^{sat}>)^2$$
$$= \tfrac{1}{4}[(125.0 - 125.04)^2 + (124.2 - 125.04)^2 + (124.7 - 125.04)^2 + (126.0 - 125.04)^2 + (125.3 - 125.04)^2] \text{ (mm Hg)}^2$$
$$= 0.4530 \text{ (mm Hg)}^2$$

Standard deviation $s_P = \sqrt{s_P{}^2} = 0.673$ mm Hg

$$\text{(b) } t_{\nu,\alpha} = \frac{\bar{P}^{sat} - <\bar{P}^{sat}>}{s_{\bar{P}^{sat}}} \overset{\text{Central Limit Theorem}}{\overbrace{=}} \frac{\bar{P}^{sat} - <P^{sat}>}{s_{P^{sat}}/\sqrt{5}} = \frac{\bar{P}^{sat} - 125.04 \text{ mm Hg}}{\left(0.673/\sqrt{5}\right) \text{ mm Hg}} = \frac{\bar{P}^{sat} - 125.04 \text{ mm Hg}}{0.301 \text{ mm Hg}}$$

For $\nu = 5 - 1 = 4$ and a 95% confidence interval (with 2 tails), $\alpha = (1 - 0.95)/2 = 0.025$.
From the *t*-table, $t_{\nu=4,\alpha=0.025} = \pm 2.776$

Therefore, the 95% confidence interval for the mean $\bar{P}^{sat}$ is given by

$$\bar{P}^{sat} = <\bar{P}^{sat}> + t_{\nu,\alpha} s_{\bar{P}^{sat}} = (125.04 \pm 2.776 * 0.301) \text{mm Hg} = (125.04 \pm 0.836) \text{mm Hg}$$

or $\bar{P}_L^{sat} = 124.20 \text{ mm Hg} < \bar{P}^{sat} < \bar{P}_U^{sat} = 125.88 \text{ mm Hg}$

Likewise, for the 99% confidence interval, $\alpha = \tfrac{1}{2}(1 - 0.99) = 0.005$
From the t-table, $t_{\nu=4,\alpha=0.025} = \pm 4.604$

Therefore, the 99% confidence interval for the mean $\bar{P}^{sat}$ is given by

$$\bar{P}^{sat} = <\bar{P}^{sat}> + t_{\nu,\alpha} s_{\bar{P}^{sat}} = (125.04 \pm 4.604 * 0.301) \text{mm Hg} = (125.04 \pm 1.386) \text{mm Hg}$$

or $\bar{P}_L^{sat} = 123.65 \text{ mm Hg} < \bar{P}^{sat} < \bar{P}_U^{sat} = 126.43 \text{ mm Hg}$

**Example 15** A newly manufactured product contains an impurity $X$ whose true mean ($\mu_X$) and true standard deviation ($\sigma_X$) in concentration level are unknown. Suppose an analysis of five random samples yields a sample mean ($<X>$) of 1.8% and a sample standard deviation ($s_X$) of 0.40% in the concentration level. Assuming that the impurity concentration level conforms to the t-distribution, determine the following:

(a) the 95% confidence interval for $\bar{X}$,
(b) the probability that $\bar{X}$ will fall outside the (1.8 ± 0.1)% range,
(c) the probability that $\bar{X}$ will exceed 2.0%, and
(d) the probability that $\bar{X}$ will fall within the 1:00-2:00% range.

**Solution**

Use the *t*-distribution function and the definition of t to relate *t* to $\bar{X}$,

$$t_{\nu,\alpha} = \frac{\bar{X} - <\bar{X}>}{s_{\bar{X}}} = \frac{\bar{X} - <\bar{X}>}{s_X/\sqrt{n}} = \frac{\bar{X} - 1.8\%}{0.40\%/\sqrt{5}} = \frac{\bar{X} - 1.8\%}{0.179\%}.$$

(a) For a 95% confidence interval, the tail area $\alpha = (1 - 0.95)/2 = 0.025$. For the degrees of freedom $\nu = 5 - 1 = 4$, and with $\alpha = 0.025$, the *t*-table gives $t_{\nu=4,\alpha=0.025} = \pm 2.776$.

Therefore, $\pm 2.776 = t_{\nu=4,\alpha=0.025} = \dfrac{\bar{X} - 1.8\%}{0.179\%}$ or the 95% confidence interval for $\bar{X}$ is given by

$\bar{X} = 1.8\% \pm 2.776*0.179\% = (1.8 \pm 0.497)\%$ or $\bar{X}_L = 1.303\% < \bar{X} < \bar{X}_U = 2.296\%$.

(b) The *t*- interval corresponding to the $\bar{X} = (1.8 \pm 0.1)\%$ interval is given by

$$t_{\nu,\alpha} = \frac{\bar{X} - <\bar{X}>}{s_X/\sqrt{n}} = \frac{(1.8 \pm 0.1)\% - 1.8\%}{0.179\%} = \pm 0.559$$

Using the *t*-distribution calculator program on http://stattrek.com/Tables/T.aspx, for $\nu = 4$ and $t = \pm 0.559$, $\alpha = 1 - 0.6970 = 0.3030$.

Thus, $P(1.7\% < \bar{X} < 1.9\%) = P(-0.559 < t < 0.559)$
$= 2*(0.500 - 0.3030)$
$= 2*0.1970$
$= 0.3940$

Therefore, $P(\bar{X}$ is outside the 1.8 ± 0.1 % range$) = P(\bar{X} < 1.7\%$ or $\bar{X} > 1.9\%)$
$= 1 - \text{Prob}(1.7\% < \bar{X} < 1.9\%)$
$= 1 - 0.3940$
$= 0.6060$

(c) For $\bar{X} > 2.0\%$ or $t > \dfrac{\bar{X} - <\bar{X}>}{s_X/\sqrt{n}} = \dfrac{2.0\% - 1.8\%}{0.179\%} = 1.118$,

$$P(\bar{X} > 2\%) = P(t_{\nu=4} > 1.118) = \int_{1.118}^{\infty} f(t)\,dt = \alpha$$

Using the *t*-distribution calculator program on http://stattrek.com/Tables/T.aspx, for $\nu = 4$, and $t = 1.118$, $\alpha = 1 - 0.8369 = 0.1631$ Therefore, $P(\bar{X} > 2\%) = 0.1631$.

(d) For $\bar{X} = 1.0\%$, $t = \dfrac{\bar{X} - <\bar{X}>}{s_X/\sqrt{n}} = \dfrac{1.0\% - 1.8\%}{0.179\%} = -4.472$

For $\bar{X} = 2.0\%$, $t_{\nu=4} = 1.118$ from part c.

$P(1.0\% < \overline{X} < 2.0\%) = P(-4.472 < t_{\nu=4} < 1.118)$

However, $P(-4.472 < t_{\nu=4} < 1.118) = P(-4.472 < t_{\nu=4} < 0) + P(0 < t_{\nu=4} < 1.118)$

$$\overset{\text{symmetry}}{=} P\left(0 < t_{\nu=4} < 4.472\right) + P\left(0 < t_{\nu=4} < 1.118\right)$$

$$= \int_0^{4.472} f(t, \nu = 4)dt + \int_0^{1.118} f(t, \nu = 4)dt$$

$$\overset{\text{t-calculator}}{=} (0.9945 - 0.5000) + (0.8369 - 0.5000)$$

$$= 0.8314$$

Alternatively, $P\left(-4.472 < t_{\nu=4} < 1.118\right) = 1 - \left(\alpha_{\nu=4, t=-4.472} + \alpha_{\nu=4, t=1.118}\right)$

$$\alpha_{\nu=4, t=-4.472} \overset{\text{symmetry}}{=} \alpha_{\nu=4, t=4.472} \overset{t-\text{calculator}}{=} 1 - 0.9945 = 0.0055$$

$$\alpha_{\nu=4, t=1.118} \overset{t-\text{calculator}}{=} 1 - 0.8369 = 0.1631$$

$\therefore\ P\left(-4.472 < t_{\nu=4} < 1.118\right) = 1 - \left(0.0055 + 0.1631\right) = 0.8314$ same as above.

Alternatively, using Table 2 and linear interpolation,

$$P\left(-4.472 < t_{\nu=4} < 1.118\right) = 1 - \left(\alpha_{\nu=4, t=-4.472} + \alpha_{\nu=4, t=1.118}\right)$$

$$\alpha_{\nu=4, t=-4.472} \overset{\text{symmetry}}{=} \alpha_{\nu=4, t=4.472} \overset{\text{linear interpolation using Table 2}}{=} 0.010 + \frac{4.472 - 3.747}{4.604 - 3.747}(0.005 - 0.010)$$

$$\alpha_{\nu=4, t=-4.472} = 0.00577$$

$$\alpha_{\nu=4, t=1.118} \overset{\text{linear interpolation using Table 2}}{=} 0.20 + \frac{1.118 - 0.941}{1.190 - 0.941}(0.15 - 0.20) = 0.1645$$

$$P\left(-4.472 < t_{\nu=4} < 1.118\right) = 1 - \left(0.00577 + 0.1645\right) = 0.8298$$

The result based on linear interpolation of Table 2 is slightly different from that using the $t$-calculator program.

**Example 16** The concentration of sulfur dioxide in the air around a coal-fired power plant has been measured six times with the following results: 320, 295, 306, 284, 315, 290 ppm. Determine (a) the sample mean and sample standard deviation, and (b) the 95% confidence interval for the average sulfur dioxide concentration. (c) Repeat part (b) if, instead of six measurements, only three measurements were made with the same sample mean and sample standard deviation as in part (a). (d) What conclusion may be made based on the result in part (c)?

**Solution**

(a) Sample mean $SO_2$ concentration is given by

$$< C_{SO_2} > = \frac{1}{n}\sum_{i=1}^{n}\left(C_{SO_2}\right)_i = \frac{1}{6}\left(320 + 295 + 306 + 284 + 315 + 290\right)\text{ppm} = 301.7\text{ ppm}$$

Sample variance of $C_{SO_2}$ is given by

$$s_{C_{SO_2}}{}^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(C_{SO_2} - < C_{SO_2} >\right)_i^2$$

$$s_{C_{SO_2}}{}^2 = \frac{1}{5}\left[\begin{array}{l}(320-301.7)^2 + (295-301.7)^2 + (306-301.7)^2 \\ +(284-301.7)^2 + (315-301.7)^2 + (290-301.7)^2\end{array}\right]\text{ppm}^2$$

$$s_{C_{SO_2}}{}^2 = 205.1\text{ ppm}^2$$

Sample standard deviation of $C_{SO_2}$ is given by $s_{C_{SO_2}}$ = SQRT($s_{C_{SO_2}}{}^2$) = ± 14.32 ppm

(b) Use *t*-statistics to determine the 95% confidence interval for the *average* sulfur dioxide concentration $\bar{C}_{SO_2}$.

For the six measurements, degrees of freedom ν = 6 – 1 = 5.

For a 95% confidence interval, $\alpha = (1 - 0.95)/2 = 0.025$.

From the t-table, for $\nu = 5$ and $\alpha = 0.025$, $t_{\nu=5,\alpha=0.025} = \pm 2.571$

$$t_{\nu=5,\alpha=0.025} = 2.571 = \frac{\bar{C}_{SO_2} - < \bar{C}_{SO_2} >}{s_{\bar{C}_{SO_2}}} = \frac{\bar{C}_{SO_2} - < C_{SO_2} >}{s_{C_{SO_2}}/\sqrt{n}} = \frac{\bar{C}_{SO_2} - 301.7\text{ ppm}}{14.32\text{ ppm}/\sqrt{6}}$$

Therefore, the 95% confidence interval for $\bar{C}_{SO_2}$ is given by

$$\bar{C}_{SO_2} = < C_{SO_2} > \pm 2.571 * s_{\bar{C}_{SO_2}} = \left(301.7 \pm 2.571 * \frac{14.32}{\sqrt{6}}\right)\text{ppm} = (301.7 \pm 15.03)\text{ ppm}$$

or 286.7 ppm < $\bar{C}_{SO_2}$ < 316.7 ppm

(c) If $n = 3$ instead of 6 and if the sample mean and sample standard deviation remain the same as in part (a), then $t_{\nu=2,\alpha=0.025} = \dfrac{\bar{C}_{SO2} - < \bar{C}_{SO_2} >}{s_{\bar{C}_{SO_2}}} = \dfrac{\bar{C}_{SO2} - < C_{SO_2} >}{s_{C_{SO_2}}/\sqrt{n}} = 4.303$

The 95% confidence interval for $\bar{C}_{SO_2}$ would then be given by

$$\bar{C}_{SO_2} = \left(301.7 \pm 4.303 * \frac{14.32}{\sqrt{3}}\right)\text{ppm} = (301.7 \pm 35.58)\text{ ppm}$$

or 266.1 ppm < $\bar{C}_{SO_2}$ < 337.3 ppm

(d) It is seen that the 95% confidence interval for $\bar{C}_{SO_2}$ based on a sample size of 3 is broader than that based on a sample size of 6, reflecting a greater uncertainty associated with a smaller sample size. The result reflects a general observation, namely, a price to pay for a smaller sample size is a broader confidence interval and a greater uncertainty in the result.